Laboratories for Universal Negatives

P versus **NP** as the existence of an experimental regime,
in the rho calculus with context-labelled bisimilarity

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Abstract

We recast the separation of **P** from **NP**, if there is one, as a statement about which experiments a process can perform on another process. The setting is the rho calculus with context-labelled transitions: a label is a context, and an observer class is a set of contexts. Three steps. First, on bounded processes bisimilarity is polynomial and trace equivalence is **coNP**-complete, and **P** = **NP** holds exactly when a polynomial-time normaliser turns trace equivalence into bisimilarity; a faithful normaliser whose output is determinate provably does not exist. Second, internalising the normaliser as a rho process $[\![\Phi]\!](x)$ that receives its input as code on x shows that everything it can do with that code is interaction: the context $[\![\Phi]\!](x) \mid x!([\,\cdot\,])$ places its hole under a lift, where communication depends on name equality, and the only uses of a received name are as a channel and under a drop. What distinguishes code from a live process is that code supports repeated experiments. Third, we sharpen the graded interaction of fuzzy rho into an apparatus for *controlled* experiments — stamping for isolation and attribution, positional interventions, a unit-grade control, pooled readout — and define an experimental regime as an access discipline, an intervention language, a readout and a cost model. The result is a classification. Black-box regimes are provably unable to certify that a bounded process cannot succeed: an adversary argument costs them 2^d intervened copies on inputs of size $O(d \log d)$. Under two stated cost hypotheses, **P** = **NP** if and only if some admissible regime certifies such universal negatives at polynomial cost. The black-box separation is an in-calculus analog of an oracle separation, so what remains of **P** ≠ **NP** in this language is entirely a statement about white-box access.

1 Introduction

This note began as a discussion of whether **P** ≠ **NP** could be independent of ZFC, and of whether forcing could be adapted to show it. The obstruction to the second question is well understood. The statement **P** ≠ **NP** is arithmetical (Π^0_2). Forcing adds no new natural numbers, so arithmetical statements are absolute between a model and its forcing extensions, and no set-theoretic forcing changes the truth value of **P** ≠ **NP**. (Shoenfield's theorem [22] carries absoluteness much further, to Σ^1_2.) Forcing survives only in its model-theoretic form, over nonstandard models of weak arithmetic [10]. Related unprovability results for circuit lower bounds in bounded arithmetic are obtained by natural-proofs arguments rather than by forcing [20]. Independence from strong theories would also be close to a positive answer. Ben-David and Halevi [7] show that if **P** ≠ **NP** is independent of PA augmented with all true Π_1 sentences

— a strong, non-recursively-enumerable theory — then **NP** has almost-polynomial algorithms. See [1] for a survey.

Rather than pursue independence directly, we ask what shape the obstruction takes when the question is posed in a calculus whose native notion of equivalence is behavioural. The answer turns out to be operational: it is a question about laboratories.

The idea, in four lines

(1) Branching time is cheap and linear time is not: on bounded processes, bisimilarity is decidable in polynomial time and trace equivalence is **coNP**-complete (§3).

(2) $\mathbf{P} = \mathbf{NP}$ if and only if a polynomial-time normaliser sends trace equivalence to bisimilarity (§4).

(3) Internalise the normaliser as a rho process $[\![\Phi]\!](x)$ receiving its input as code on x. Everything it does with that code is an experiment, and code differs from a live process in admitting repeated experiments (§5–§6).

(4) $\mathbf{P} \neq \mathbf{NP}$ if and only if no admissible experimental regime lets $[\![\Phi]\!](x)$ certify a universal negative at polynomial cost. Black-box regimes provably cannot; white-box regimes are exactly as hard as **P** versus **NP** (§7–§8).

Contributions

(i) A bisimulation-native statement of **P** versus **NP** (Theorem 4.2), with an unconditional lower bound for faithful determinate normalisers (Proposition 4.3).

(ii) The observation that a reflective normaliser is a context whose hole lies under a lift, and that in such a context every operation on the input is an interaction (Lemma 5.1); the access tiers that follow are distinguished by the structural rules they permit (§6).

(iii) An apparatus for controlled experiments built from the graded interaction of fuzzy rho [9], with the two conditions — inert signatures and equivariance — under which a stamped copy is a faithful copy, and a counterexample showing that flat stamping is not equivariant (Example 7.4).

(iv) A definition of experimental regime, an exponential lower bound for black-box regimes that is unconditional in the number of experiments (Theorem 8.5), and a characterisation of $\mathbf{P} = \mathbf{NP}$ by the existence of an admissible regime (Theorem 8.9), relative to two stated cost hypotheses.

(v) A heuristic map from the known barriers to features of the apparatus (§9).

Appendix A records the status of each result. Several are proved; some are proved relative to stated hypotheses; some are sketched; some are conjectures, and are labelled as such. Nothing here claims progress on **P** versus **NP** itself. The contribution is a reformulation whose parts are the parts of a laboratory, together with the place in that laboratory where the known barriers sit.

2 The rho calculus and context-labelled bisimilarity

Definition 2.1 (Syntax)**.** Processes P and names x are given by mutual recursion:

$$P \;::=\; \mathbf{0} \;\mid\; P_1 \mid P_2 \;\mid\; x!(P) \;\mid\; \mathsf{for}(y_1 \leftarrow x_1 \,\&\, \cdots \,\&\, y_n \leftarrow x_n)P \;\mid\; *x \qquad\qquad x \;::=\; @P$$

$x!(P)$ is the *lift*: it quotes P and outputs the name $@P$ on x. The input binds $y_1, \ldots, y_n$ in P; for $n = 1$ we write $\mathsf{for}(\, y \leftarrow x\,)P$, and for $n > 1$ it is a *join*. $*x$ is the *drop*. We write _ for a bound name that does not occur. A process is *closed* if no variable bound by an input occurs free in it; every quoted name in a closed process is then a constant.

This is the calculus of [14] with joins, in the notation of [12]. A name is in *normal form* if no quote in it immediately encloses a drop. Rewriting $@(*x)$ to x throughout gives a normal form $\equiv_{\mathcal{N}}$-equal to any given name. Name equivalence is decidable in polynomial time, being equality of unordered trees after normalisation.

Definition 2.2 (Name equivalence, structural congruence, reduction)**.** As in [12]: $\equiv_{\mathcal{N}}$ is the least equivalence on names with $@P_1 \equiv_{\mathcal{N}} @P_2$ whenever $P_1 \equiv P_2$, and $@(*x_1) \equiv_{\mathcal{N}} x_2$ whenever $x_1 \equiv_{\mathcal{N}} x_2$; $\equiv$ is the least congruence containing α-equivalence and the abelian monoid laws for $\mid$ with unit $\mathbf{0}$, names being compared up to $\equiv_{\mathcal{N}}$. Reduction is

$$\frac{x_i \equiv_{\mathcal{N}} x'_i \quad (1 \le i \le n)}{\mathsf{for}(y_1 \leftarrow x_1 \,\&\, \cdots \,\&\, y_n \leftarrow x_n)P \;\mid\; x'_1!(Q_1) \mid \cdots \mid x'_n!(Q_n) \;\to\; P\{@Q_1/y_1, \ldots, @Q_n/y_n\}}$$

closed under $\mid$ and $\equiv$. Substitution replaces all names $\equiv_{\mathcal{N}}$-equal to the variable, reaches the object of a lift, does not reach beneath a quote, and performs the drop: $(*x)\{@P/y\} = P$ when $x \equiv_{\mathcal{N}} y$.

The duplicator $\mathsf{D}(x) \triangleq \mathsf{for}(\, y \leftarrow x\,)(*y \mid x!(*y))$ satisfies $x!(Q) \mid \mathsf{D}(x) \to Q \mid x!(Q)$; it is the source of contraction on names.

2.1 Context-labelled transitions

We do not use barbs. Following [12], observation is by contexts.

Definition 2.3 (Context-labelled transitions and bisimilarity)**.** Write $P \xrightarrow{C} P'$ when $C[P] \to P'$, and $P \overset{C}{\Longrightarrow} P'$ when $C[P] \to^* P'$. For an *observer class* $\mathcal{D} \subseteq \mathcal{C}$, a symmetric relation $\mathcal{R}$ is a $\mathcal{D}$-bisimulation if $P \,\mathcal{R}\, Q$ and $P \xrightarrow{C} P'$ with $C \in \mathcal{D}$ imply $Q \overset{C}{\Longrightarrow} Q'$ with $P' \,\mathcal{R}\, Q'$; $\approx_{\mathcal{D}}$ is the largest such.

Definition 2.4 (Traces)**.** An $\mathcal{D}$-trace of P is a sequence $C_1 \cdots C_k$ of contexts in $\mathcal{D}$ with $P \overset{C_1}{\Longrightarrow} P_1 \overset{C_2}{\Longrightarrow} \cdots \overset{C_k}{\Longrightarrow} P_k$. $P \equiv^{\mathcal{D}}_{\mathrm{tr}} Q$ when P and Q have the same $\mathcal{D}$-traces. Plainly $\approx_{\mathcal{D}} \subseteq \equiv^{\mathcal{D}}_{\mathrm{tr}}$.

Definition 2.5 (Success)**.** Fix a name ok and let $C_{\mathsf{ok}} = [\,\cdot\,] \mid \mathsf{for}(\,_ \leftarrow \mathsf{ok}\,)\mathbf{0}$. P *can succeed* if $P \overset{[\,\cdot\,]}{\Longrightarrow} P' \xrightarrow{C_{\mathsf{ok}}} P''$ for some P', P'' in which the displayed step consumes an output on ok.

Success is thus a context-labelled transition, not a barb. The whole of this note is about the choice of $\mathcal{D}$: an experimental regime will be, among other things, an observer class.

Remark 2.6 (Holes under a lift)*.* The companion paper [24] shows that in rho, context-labelled bisimilarity is a congruence for parallel composition, lift, input and drop-of-a-quote when observations are suitably restricted, and fails to be one for contexts with the hole under a quote, because communication depends on name equality rather than on bisimilarity. The contexts that matter in §5 are exactly of that kind.

2.2 Cost

Cost-accounted rho charges phlogiston for communication. We need only two properties of the charge, stated as hypotheses because they are properties of a cost model, not of the calculus.

Hypothesis 2.7 (Simulation)**.** There is a polynomial q such that, for every deterministic Turing machine M, there is a rho process $R_M(x)$ which, given an input word w as data on x, runs determinately to a representation of $M(w)$ at phlogiston at most $q(t_M(w) + |w|)$, where $t_M(w)$ is the running time.

Hypothesis 2.8 (Accounting)**.** Each communication — including the graded communications of §7 and the probe interactions of §6 — is charged at least 1 plus the total size of the transmitted terms. Consequently a run of phlogiston c from a term of size n can be simulated by a deterministic Turing machine in time polynomial in $c + n$.

Hypothesis 2.8 excludes the size explosion that free duplication would otherwise allow: a process that repeatedly receives Q and sends $Q \mid Q$ pays for every copy it makes.

3 Bounded processes

Two presentations of bounded behaviour are needed: an explicit one, on which the bisimulation analog is stated, and a succinct one, on which experiments are performed. They are polynomially inter-reducible through **coNP**-completeness.

3.1 Explicit presentation

Definition 3.1. An *explicit bounded process* is a finite acyclic labelled transition system $G = (S, \Sigma, \to, s_0)$, given as a list of transitions; $|G|$ is the length of that list. G is *determinate* if every state has at most one a-successor for each $a \in \Sigma$. $\sim$ is strong bisimilarity and $\equiv_{\mathrm{tr}}$ trace equivalence.

Construction 3.2 (Encoding)**.** Fix pairwise distinct closed names $\mathsf{go}_s, \mathsf{l}_s$ for $s \in S$ and a for $a \in \Sigma$, chosen injectively (for instance by quoting numerals). Put

$$\mathcal{E}(s) \;=\; \mathsf{l}_s!(\mathbf{0}) \;\mid \prod_{(s,a,t)\in\to} \mathsf{for}(_ \leftarrow a \,\&\, _ \leftarrow \mathsf{l}_s)\, \mathsf{go}_t!(\mathbf{0}),$$

$$\mathcal{E}(G) \;=\; \mathsf{go}_{s_0}!(\mathbf{0}) \;\mid \prod_{s\in S} \mathsf{for}(_ \leftarrow \mathsf{go}_s)\mathcal{E}(s).$$

The lock l_s lets exactly one outgoing transition of s fire; a state shared by several paths is encoded once and triggered at most once per run, because G is acyclic. $|\mathcal{E}(G)| = O(|G|)$. The observer class is $\mathcal{D}_\Sigma = \{\, [\,\cdot\,] \mid a!(\mathbf{0}) : a \in \Sigma \,\}$.

Communications on go-names are *administrative*; a join consuming an offered $a!(\mathbf{0})$ is *principal*, in the sense of the phase separation of [12]. Administrative reductions are deterministic (each go_t has one listener) and terminate. A losing branch of s is left as a join on l_s, which is never sent again and which no context in $\mathcal{D}_\Sigma$ can name, so it is inert.

Proposition 3.3 (Adequacy of the encoding)**.** *Taking transitions to be principal steps followed by administrative normalisation, $s \sim t$ iff $\mathcal{E}(s) \approx_{\mathcal{D}_\Sigma} \mathcal{E}(t)$, and $s \equiv_{\mathrm{tr}} t$ iff $\mathcal{E}(s) \equiv_{\mathrm{tr}}^{\mathcal{D}_\Sigma} \mathcal{E}(t)$.*

Proof sketch. After administrative normalisation, a term reachable from $\mathcal{E}(G)$ is $\mathcal{E}(s) \mid R$ with R a parallel composition of inert joins and spent triggers. A principal step on $a!(\mathbf{0})$ is available from it exactly when $s \xrightarrow{a} t$ for some t, and leads, after normalisation, to $\mathcal{E}(t) \mid R'$. The residue R contributes no $\mathcal{D}_\Sigma$-transitions. Both equivalences then transfer along the correspondence. □

Remark 3.4. The restriction to $\mathcal{D}_\Sigma$ matters. An arbitrary rho context can name l_s, since it is computed by the encoding, and so revive a losing branch, much as an arbitrary context separates encodings in the necessity result of [12]. Reviving a branch uses knowledge of the term, which is a white-box operation in the sense of §6.

Theorem 3.5 (Known complexity)**.** *On explicit bounded processes:*

(a) $\sim$ *is decidable in polynomial time [15]; on finite transition systems in general it is* **P***-complete [6];*

(b) $\equiv_{\mathrm{tr}}$ *is* **coNP***-complete.*

Proof of (b), sketch. Inequivalence is in **NP**: guess a trace, of length at most the depth, present in one system and absent from the other; membership of a given trace is decided by tracking the set of reachable states. Hardness: equivalence of regular expressions without star is **coNP**-hard [23]; such an expression converts in polynomial time to an acyclic automaton, and marking acceptance by a final ok-transition and trimming states that cannot reach acceptance turns language equivalence into trace equivalence. □

3.2 Succinct presentation

Definition 3.6. A process is *flat* if it is closed, contains no drop (not even beneath a quote), and no bound name occurs in subject position. In a flat process each communication consumes an input prefix not under a lift, and none is created, since no transmitted term is ever run. So every run of a flat P has length at most $|P|$. Let Absence be the problem: given a flat P, decide that P *cannot* succeed.

Proposition 3.7. Absence *is* **coNP***-complete.*

Proof. It is in **coNP**, since a successful run has length at most $|P|$ and is a certificate of success, checkable in polynomial time because name equivalence is. For hardness, reduce from CNF unsatisfiability. Given variables $v_1, \ldots, v_d$ and clauses $c_1, \ldots, c_m$, fix pairwise distinct closed names $b_i, t_{ij}, f_{ij}, e_j, \mathsf{ok}$, and let

$$\mathit{Race}_i \;=\; b_i!(\mathbf{0}) \;\mid\; \mathsf{for}(_ \leftarrow b_i)\textstyle\prod_{j:v_i \in c_j} t_{ij}!(\mathbf{0}) \;\mid\; \mathsf{for}(_ \leftarrow b_i)\textstyle\prod_{j:\neg v_i \in c_j} f_{ij}!(\mathbf{0}),$$

let Clause_j be the parallel composition of $\mathsf{for}(_ \leftarrow t_{ij})e_j!(\mathbf{0})$ for each positive occurrence of v_i in c_j and $\mathsf{for}(_ \leftarrow f_{ij})e_j!(\mathbf{0})$ for each negative one, and put

$$P_\varphi \;=\; \prod_i \mathit{Race}_i \;\mid\; \prod_j \mathit{Clause}_j \;\mid\; \mathsf{for}(_ \leftarrow e_1 \,\&\, \cdots \,\&\, _ \leftarrow e_m)\,\mathsf{ok}!(\mathbf{0}).$$

Each race is won by exactly one listener, which fixes a truth value for v_i; a clause signals e_j only if one of its literals was made true; the final join fires only if every clause signalled. A satisfying assignment yields a successful schedule and conversely, so P_φ can succeed iff φ is satisfiable. P_φ is flat and of size polynomial in φ. □

The races are where scheduling nondeterminism lives. A *resolution* of a flat process assigns a winner to each race. The point of §7 is to make resolutions the experimenter's choice rather than the scheduler's.

4 Normalisers

Definition 4.1. A *normaliser* is a polynomial-time map Φ on explicit bounded processes such that for all G, H

$$G \equiv_{\mathrm{tr}} H \quad \Longleftrightarrow \quad \Phi(G) \sim \Phi(H).$$

It is *faithful* if moreover $\Phi(G) \equiv_{\mathrm{tr}} G$, and *determinate* if every $\Phi(G)$ is determinate.

Theorem 4.2. $\mathbf{P} = \mathbf{NP}$ *if and only if a normaliser exists, and then a faithful one exists.*

Proof. ($\Leftarrow$) Given Φ, decide $G \equiv_{\mathrm{tr}} H$ by computing $\Phi(G)$ and $\Phi(H)$ and testing $\sim$ (Theorem 3.5(a)). This puts a **coNP**-complete problem in $\mathbf{P}$, so $\mathbf{coNP} = \mathbf{P}$ and hence $\mathbf{NP} = \mathbf{P}$.

($\Rightarrow$) If $\mathbf{P} = \mathbf{NP}$ the polynomial hierarchy collapses to $\mathbf{P}$. Let $m(G)$ be the least size of an explicit process trace-equivalent to G, and let $\Phi(G)$ be the lexicographically least explicit process G', encoded as a list of transitions over states numbered $0, \ldots, m$ and over the labels occurring in traces of G, with $|G'| = m(G)$ and $G' \equiv_{\mathrm{tr}} G$. Both depend only on the trace class of G. Since $m(G) \leq |G|$, $m(G)$ is found by asking, for each $m \leq |G|$, whether some G' of size m is trace-equivalent to G. The bits of $\Phi(G)$ are then found one at a time by asking whether some such G' extends a given prefix. Each question has the form: there is a G' such that for all traces up to the depth bound, membership agrees. That is Σ_2^p, hence polynomial-time. (This is the standard fact that $\mathbf{P} = \mathbf{NP}$ yields polynomial-time canonical forms for **coNP** equivalence relations.) Then $\Phi(G) \equiv_{\mathrm{tr}} G$, so Φ is faithful. If $G \equiv_{\mathrm{tr}} H$ then $\Phi(G) = \Phi(H)$, hence $\Phi(G) \sim \Phi(H)$. Conversely $\Phi(G) \sim \Phi(H)$ implies $\Phi(G) \equiv_{\mathrm{tr}} \Phi(H)$, hence $G \equiv_{\mathrm{tr}} H$. □

Proposition 4.3 (Faithful determinate normalisers do not exist)**.** *There are explicit bounded processes G_n with $|G_n| = O(n^2)$ such that every determinate H with $H \equiv_{\mathrm{tr}} G_n$ has at least 2^n states. In particular no faithful determinate normaliser runs in polynomial time, unconditionally.*

Proof. Let $L_n = \{\, uv \in \{0,1\}^{2n} : u \neq v,\ |u| = |v| = n \,\}$, and let G_n have the traces of the words of L_n each followed by ok. G_n guesses a position $i \leq n$, records the bit read there, counts n further positions, checks that the bit there differs, reads to length $2n$ and emits ok. Its states are indexed by a position, a guessed index and a bit, so there are $O(n^2)$ of them, and it is acyclic. Let H be determinate with $H \equiv_{\mathrm{tr}} G_n$. Each $u \in \{0,1\}^n$ is a trace of H, since uu' is in L_n for any $u' \neq u$, and so leads to a unique state h_u. For $u \neq u'$ the trace $u'\mathsf{ok}$ is available from h_u but not from $h_{u'}$, since $u'u' \notin L_n$. So the h_u are pairwise distinct. This is the classical blow-up of determinisation [16, 17], specialised to finite languages. □

Faithfulness matters. Without it a determinate normaliser exists iff $\mathbf{P} = \mathbf{NP}$: if $\mathbf{P} = \mathbf{NP}$, map G to the straight-line system spelling out the bits of the canonical form of Theorem 4.2. Straight lines are determinate, and two of them are bisimilar exactly when the canonical forms are equal.

Remark 4.4 (The ladder). There are three grades of normaliser. A faithful *determinate* normaliser is excluded unconditionally (Proposition 4.3). An *arbitrary* polynomial-time normaliser exists iff $\mathbf{P} = \mathbf{NP}$ (Theorem 4.2). Between them sit *compositional* normalisers, which act on the encoded term $\mathcal{E}(G)$ in a syntax-directed way and commute with its operators, parallel composition and prefixing. For these we expect exponential lower bounds of the subset-construction kind (Conjecture 4.5), much as monotone circuits admit lower bounds that general circuits have so far resisted.

Conjecture 4.5. *No faithful compositional normaliser runs in polynomial time.*

5 Internalising the normaliser

Write the internalised normaliser as $[\![\Phi]\!](x)$: a fixed rho process, independent of its input, with a free name x on which it expects that input. The configuration is

$$[\![\Phi]\!](x) \;\mid\; x!(P) \;\;\to^* \;\; P'.$$

In context-labelled terms the experimenter *is* the context $C_\Phi = [\![\Phi]\!](x) \mid x!([\,\cdot\,])$, and its hole lies under a lift. By Remark 2.6 this is the position where $\approx$ fails to be a congruence. So the requirement that $P \approx Q$ imply $C_\Phi[P] \approx C_\Phi[Q]$, and *a fortiori* the requirement that $P \equiv_{\mathrm{tr}} Q$ imply it, is not free: it is the burden the normaliser carries. The live alternative, $[\![\Phi]\!] \mid [\,\cdot\,]$, gets invariance under $\approx$ for free from congruence and loses faithfulness instead (Example 5.2).

Lemma 5.1 (What a received name is good for)**.** *In* $\mathsf{for}(\, y \leftarrow x\,)R$*, every free occurrence of* y *in* R *is either the subject of an input or output or the argument of a drop. Consequently, once* $[\![\Phi]\!](x)$ *has received* $@P$*, it can*

(i) run P*, by a drop; or*

(ii) use $@P$ *as a channel, which matters only to processes that also hold a name* $\equiv_{\mathcal{N}} @P$*, and so amounts to testing* $@P$ *for name equality against names it or its environment constructs; or*

(iii) forward it inside a lifted term, which defers (i) or (ii) to the receiver.

Proof. By the grammar, names occur only as subjects of input and output, as binders, under drop, and beneath quotes; occurrences beneath a quote are not free, since substitution does not reach there. Communication on a name fires only when the two subjects are $\equiv_{\mathcal{N}}$-equal. □

Pure rho has no third operation, no deconstructor. Any test on the syntax of P is an interaction: either running P against a context, or a communication whose firing condition is a name equality the experimenter arranged. So the quoted reading reduces, internally, to possibly iterated interaction with P. What separates it from the live reading is *resource*: interaction with a live P can use P up, while a name can be dropped as often as the experimenter pays for.

Example 5.2 (Why one experiment is not enough). Let $G = a.(b + c)$ and $H = a.b + a.c$, so $G \equiv_{\mathrm{tr}} H$ and $G \not\sim H$. A context interacting with the live $\mathcal{E}(H)$ learns nothing until it offers a, and in $\mathcal{E}(H)$ the very join that consumes the offer also chooses the branch. Afterwards the continuation can do b or c, not both. The losing branch survives only as a dead join on a lock, which no observer in $\mathcal{D}_\Sigma$ can revive. A faithful normal form must have both traces ab and ac, and a single linear experiment on $\mathcal{E}(H)$ cannot recover the one it did not see. In a calculus with summation the losing branch is discarded outright.

6 Access tiers and structural rules

Quotation gives the input the exponential of linear logic: $@P$ behaves as $!P$, the drop as dereliction, the duplicator as contraction. Access tiers are distinguished by which of these rules the experimenter may use on the received name, and by whether it may inspect the term.

Definition 6.1 (Tiers)**.** (a) *Linear*: the received name is dropped at most once, as in the exactly-once input $\mathsf{for}(y \multimap x)P$ of the linear fragment. No contraction.

(b) *Replicable black-box*: the received name is used only under drop, and is forwarded only to receivers that likewise use it only under drop (*drop-only* usage). The experimenter's contexts mention, of the copy's names, only those in a declared public interface (here ok and any declared intervention ports). Contraction is allowed; inspection and name guessing are not.

(c) *White-box*: as (b), and in addition the experimenter may apply the probes of the observer extension [13] to the received name itself — to the quoted code, which does not run while it is read. The probes make each term constructor observable by interaction: an atom Ask_C meeting $C(t_1, \ldots, t_n)$ yields its arguments, and $\mathsf{Get}_{C,i}$ selects the i-th. Under the capability-guarded refinement of that construction, holding x is what licenses inspection.

Name guessing — use (ii) of Lemma 5.1 against names the experimenter constructs — is excluded from tier (b) deliberately. It is a syntactic test, and it belongs with inspection.

Conjecture 6.2 (Drop-only contexts are congruent)**.** *If the hole of C lies under a lift and the corresponding name is used drop-only, then $P \approx_{\mathcal{D}} Q$ implies $C[P] \approx_{\mathcal{D}} C[Q]$ for suitable $\mathcal{D}$. That is, drop-only usage makes a quote transparent, and tier (b) is an admissible observer class in the sense of [24].*

7 Controlled experiments: sharpening fuzzy rho

Repetition alone yields only existential evidence. If a copy of P succeeds, P can succeed. If it fails, the scheduler may simply have resolved a race the other way. A controlled experiment needs isolation, attribution, fidelity, intervention, a control and a readout. We take each from the graded interaction of fuzzy rho [9], in which the transmitted term is modulated by a grade comparing the two interaction surfaces:

$$\mathsf{for}(\, y \leftarrow x\,)R \;\mid\; u!(Q) \;\to\; R\{@((u \circ x) \star Q)/y\}, \qquad (u \circ x) \star Q = Q\{@(\overline{u} \mid \overline{x} \mid \overline{z})/z : z \in \mathrm{fn}(Q)\},$$

where $\overline{z}$ is the process quoted by z. This is the *accumulator* model; the *attenuator* replaces $\mid$ by the multiset meet $\wedge$.

7.1 Isolation and attribution: the stamp

Definition 7.1 (Stamp)**.** A *signature* is a closed process S having, as a $\mid$-prime component, an inert input $K_S = \mathsf{for}(\,_ \leftarrow k_S\,)\mathbf{0}$ on an unforgeable name k_S that occurs nowhere else. The *stamp* θ_S sends each name z in normal form to $@(S \mid \overline{z})$, and $S \star P$ applies θ_S to the names of P, taken in normal form.

Lemma 7.2 (Stamps are injective)**.** $\theta_S(z_1) \equiv_{\mathcal{N}} \theta_S(z_2)$ *implies* $z_1 \equiv_{\mathcal{N}} z_2$.

Proof. $S \mid \overline{z}$ has the input K_S as a component, so it is never $\equiv$ to a lone drop, and only the first clause of $\equiv_{\mathcal{N}}$ applies: $\theta_S(z_1) \equiv_{\mathcal{N}} \theta_S(z_2)$ means $S \mid \overline{z_1} \equiv S \mid \overline{z_2}$. Up to $\equiv$ a process is a finite multiset of $\mid$-prime components, and the free commutative monoid is cancellative, so $\overline{z_1} \equiv \overline{z_2}$, and $z_1 \equiv_{\mathcal{N}} z_2$ since both are in normal form. □

Lemma 7.3 (Distinct signatures are disjoint)**.** *Let S_1, S_2 be signatures with $k_{S_1} \neq k_{S_2}$, and z_1, z_2 names of an input P that mentions neither k_{S_i}. Then $\theta_{S_1}(z_1) \not\equiv_{\mathcal{N}} \theta_{S_2}(z_2)$.*

Proof. If $S_1 \mid \overline{z_1} \equiv S_2 \mid \overline{z_2}$, the prime component K_{S_1} occurs on the right, hence in S_2 or in $\overline{z_2}$. Neither mentions k_{S_1}. □

Run k copies with signatures $S_1, \ldots, S_k$ whose unforgeable names k_{S_i} are distinct, as the content-addressed channels of cost-accounted rho provide. Then by Lemmas 7.2 and 7.3 the copies share no channel, so they cannot interfere with one another, and every observation carries the signature of the copy that made it. The experimenter knows each S_i, so it can compute the stamped interface $\theta_{S_i}(\mathsf{ok})$ on which to listen.

7.2 Fidelity: inertness and equivariance

A stamped copy is useful only if it is a faithful copy. Two things can go wrong.

Signatures run under drop. If P contains $*z$ for a free z, the copy contains $*\theta_S(z)$, which runs $S \mid \overline{z}$. The signature must therefore be *inert*: built, for instance, from inputs on unforgeable names that nothing ever sends on.

Names synthesised at runtime escape the stamp. Flat substitution stamps the names P has at the moment it is sent, not the names it computes while it runs.

Example 7.4 (Flat stamping is not equivariant). Let $P = a!(\mathbf{0}) \mid \mathsf{for}(\, y \leftarrow a\,)y!(\mathbf{0}) \mid \mathsf{for}(\, _ \leftarrow @\mathbf{0}\,)\mathsf{ok}!(\mathbf{0})$. Then P receives $y := @\mathbf{0}$, outputs on $@\mathbf{0}$, and succeeds. In $S \star P$, for any signature S, the names a, ok and $@\mathbf{0}$ are stamped, but the transmitted name $@\mathbf{0}$ is synthesised from the object of the lift, which has no names, and so is not. The copy outputs on $@\mathbf{0}$ while its listener waits on $\theta_S(@\mathbf{0}) \equiv_{\mathcal{N}} @S$. The copy cannot succeed.

Definition 7.5 (Experimental grading)**.** A grading $S \star (-)$ is *experimental* if
(E1) its signatures are signatures in the sense above, pairwise distinct across copies;
(E2) its signatures are inert;
(E3) it is *equivariant*: $P \to P'$ implies $S \star P \to S \star P'$, and every reduction of $S \star P$ is of this form.

(E3) is the dynamic form of the binder transport law of [9], $a \star [y]P = [y](a' \star P)$. It says the copy runs *inside* the stamp, so that every name it ever creates is stamped.

Proposition 7.6. *If $S\star(-)$ is experimental, the relation $\{(P', S\star P')\}$ is a bisimulation between the reduction systems of P and of $S\star P$, with observations on z matched to observations on $\theta_S(z)$. In particular $S \star P$ can succeed at $\theta_S(\mathsf{ok})$ iff P can succeed.*

Proof. Immediate from (E3), with (E2) ensuring that the dropped signature contributes no transitions. □

Flat processes (Definition 3.6) synthesise no names that are ever used as channels and contain no drops, so flat stamping is experimental on them. Realising (E3) for general processes needs a namespace-relative evaluation of the copy; this is open (§11).

7.3 Intervention, control, dose and readout

Intervention. Uniform stamping isolates a copy but does not steer it. To control a variable we need the analog of Pearl's do(·) [18]: resolve a chosen race to a chosen branch. Stamping a single *occurrence* does this. In $z!(A) \mid \mathsf{for}(\, y \leftarrow z\,)B \mid \mathsf{for}(\, y \leftarrow z\,)C$, stamping the subject of the third component with an inert tag cuts that edge, and the race resolves to B. Occurrence-level stamping needs addresses into the term, which the probes $\mathsf{Get}_{C,i}$ supply, so positional intervention is a white-box operation. A *full intervention* τ resolves every race. Under a full intervention a flat process runs determinately, up to the interleaving of independent steps.

Control. Classical rho is the case $u = x = @\mathbf{0}$ of the graded comm [9]. A copy sent at unit grade is the untreated baseline.

Dose. The order axiom of [9] — a larger match grade transmits at least as much information — is, for partial or soft interventions, a monotone dose–response law. That law is what licenses inference from graded treatments.

Readout. The attenuator $u \circ x = @(\overline{u} \wedge \overline{x})$ can collapse distinct channels onto one shared stamped name. Collapsing the success channels of a pool of copies onto one name gives a single readout that reports whether *any* member succeeded. A negative pooled readout certifies failure for every member, provided the collapse sends nothing else to that name. Pooling saves readouts, not runs: each copy still realises one resolution.

Remark 7.7 (What failure now means). Under a full intervention τ, failure is conclusive *for* τ. The controlled apparatus therefore converts demonic scheduling into the experimenter's choice. Success of P is the existence of a τ under which the copy succeeds — an **NP** witness, now chosen rather than granted. Failure of P is failure under every τ. The **coNP** half becomes a problem of experimental design: can polynomially many interventions, possibly pooled and graded, *cover* the space of resolutions?

8 Experimental regimes

Definition 8.1 (Regime)**.** An *experimental regime* is a tuple $\mathcal{R} = (A, I, O, c)$ where

- A is an access tier (Definition 6.1);
- I is an intervention language: a set of partial resolutions the experimenter may impose, each realised by an experimental grading;
- O is a readout discipline: single, pooled or graded;
- c is the cost model, phlogiston satisfying Hypotheses 2.7 and 2.8.

A *copy* is released by a graded communication that applies its signature and its intervention; we identify the copy with that communication. An *experiment* is a finite pool of copies, run against test contexts allowed by A, with a readout in O. $\mathcal{R}$ is *admissible* if, on every flat input, every experiment has an outcome independent of the scheduler. $[\![\Phi]\!](x)$ *exploits* $\mathcal{R}$ for Absence if, for every flat P, the configuration $[\![\Phi]\!](x) \mid x!(P)$ performs only experiments of $\mathcal{R}$, runs at phlogiston polynomial in $|P|$, and reports, independently of the scheduler, whether P can succeed. (Given this last clause, admissibility is needed only of the experiments actually performed; regimes whose interventions are partial are rarely admissible, and Theorem 8.5 does not assume admissibility.)

Lemma 8.2 (Copies cost)**.** *Under Hypothesis 2.8, a run of phlogiston c releases at most c copies.*

Proof. Each copy is its own graded communication, charged at least 1. □

Remark 8.3. Unstamped duplicates are cheaper. One communication can deliver $@P$ to a body containing many drops. But duplicates share their channels, so they are neither isolated nor attributable, and none of them carries an intervention. In the black-box model below an unintervened duplicate is resolved by the scheduler, so it contributes nothing that an adversary cannot neutralise.

8.1 Black-box regimes cannot certify universal negatives

Definition 8.4 (Black-box regime)**.** $\mathcal{R}$ is *black-box* if its tier is (a) or (b) and the outcome of each experiment on P is determined by the experiment together with the success or failure, at the stamped public interface, of each copy under its intervention. Interventions are drawn from a public intervention language over declared race ports. These are granted as a primitive, which is more than a black-box experimenter could otherwise do, since positional intervention needs addresses; granting them only strengthens the lower bound below. Races left unresolved by an intervention are resolved by a scheduler outside the experimenter's control, and correctness is required under every scheduler.

This definition is generous to the experimenter. It allows full interventions without any inspection, and it allows pooled readouts and any graded readout that is a function of the per-copy outcomes. It excludes graded readouts that aggregate over the resolutions of a single copy rather than taking the one the scheduler chose; those are the subject of Conjecture 9.1.

Theorem 8.5 (Black-box lower bound)**.** *For each d there are flat processes $P_\perp$ and P_σ for $\sigma \in \{0,1\}^d$, all of size $O(d)$, such that $P_\perp$ cannot succeed, each P_σ can, and every $[\![\Phi]\!](x)$ that, using a black-box regime, reports correctly and independently of the scheduler on all of them releases at least 2^d intervened copies on input $P_\perp$. Under Hypothesis 2.8 its phlogiston on $P_\perp$ is at least 2^d.*

Proof. Let $Race_i = b_i!(\mathbf{0}) \mid \mathsf{for}(_ \leftarrow b_i\,)e_i^0!(\mathbf{0}) \mid \mathsf{for}(_ \leftarrow b_i\,)e_i^1!(\mathbf{0})$, with race port r_i naming the race on b_i. Put

$$P_\sigma = \prod_{i \le d} Race_i \;\mid\; \mathsf{for}(_ \leftarrow e_1^{\sigma_1} \,\&\, \cdots \,\&\, _ \leftarrow e_d^{\sigma_d})\,\mathsf{ok}!(\mathbf{0}),$$

$$P_\perp = \prod_{i \le d} Race_i \;\mid\; \mathsf{for}(_ \leftarrow e_1^0 \,\&\, \cdots \,\&\, _ \leftarrow e_d^0 \,\&\, _ \leftarrow z)\,\mathsf{ok}!(\mathbf{0})$$

with z a name nothing sends on, and all names pairwise distinct. (With names as quoted numerals the size is $O(d \log d)$.) Then P_σ succeeds exactly under the resolution σ, and $P_\perp$ never succeeds.

Consider a copy with intervention τ (a partial assignment of race outcomes). On $P_\perp$ it fails. On P_σ it fails unless τ is the full assignment σ: if τ disagrees with σ somewhere, the final join cannot fire; if τ leaves some race unresolved, a scheduler that resolves it against σ makes the copy fail. Fix such a scheduler. By Definition 8.4, the outcome of every experiment on P_σ then coincides with its outcome on $P_\perp$, unless the experiment contains a copy with full intervention σ.

Run $[\![\Phi]\!](x)$ on $P_\perp$ under the same scheduler. It is correct, so it reports failure. Let T be the set of full interventions it ever applied to a copy. If some $\sigma \notin T$, then on P_σ, under the scheduler above, every experiment returns what it returned on $P_\perp$. By induction on the run, $[\![\Phi]\!](x)$ makes the same adaptive choices and also reports failure, which is wrong. Hence $|T| = 2^d$. So $[\![\Phi]\!](x)$ released at least 2^d intervened copies on $P_\perp$, and by Lemma 8.2 its phlogiston is at least 2^d. □

Corollary 8.6. *No black-box regime, admissible or not, is exploitable for* Absence. *In particular a linear regime, which admits one copy, is not.*

Remark 8.7 (Why this says nothing about **P** versus **NP**). In a white-box regime the family is trivial. $[\![\Phi]\!]$ reads the final join and sees either the conjunct on z or the hidden σ. The lower bound is purely about access. It is a query lower bound for unstructured search, the in-calculus analog of the *separating* half of the Baker–Gill–Solovay theorem [5]: a black-box experimenter computes with P as an oracle, and an unconditional separation is available because the oracle hides its structure. We give no analog of the *collapsing* half, so the correspondence with the relativisation barrier is an analogy, not a theorem.

Remark 8.8 (Graded and quantum readouts). If a graded regime allowed amplitude-weighted superpositions of interventions, as in the complex-weighted GSLTs, we would expect the bound to weaken to $\Omega(2^{d/2})$ and no further, by the Bennett–Bernstein–Brassard–Vazirani argument [8]. Making this precise requires a quantum semantics of the graded comm.

8.2 White-box regimes are as hard as P versus NP

Theorem 8.9. *Assume Hypotheses 2.7 and 2.8, and that reading a term of size n by probes costs phlogiston polynomial in n. Then* $\mathbf{P} = \mathbf{NP}$ *if and only if some admissible regime is exploitable for* Absence. *When it is, a white-box regime is.*

Proof. ($\Rightarrow$) If $\mathbf{P} = \mathbf{NP}$ then Absence $\in \mathbf{P}$ by Proposition 3.7, decided by some machine M. In a white-box regime $[\![\Phi]\!](x)$ receives $@P$, reads the quoted code by probes into a data representation at polynomial cost, and runs R_M on it (Hypothesis 2.7). No copy of P ever runs, so its report does not depend on the scheduler.

($\Leftarrow$) Suppose $[\![\Phi]\!](x)$ exploits a regime $\mathcal{R}$. Its report is independent of the scheduler, so simulating any one schedule computes it, and the whole configuration runs at polynomial phlogiston. By Hypothesis 2.8 a deterministic machine simulates it in polynomial time, so Absence $\in \mathbf{P}$ and hence $\mathbf{coNP} = \mathbf{P} = \mathbf{NP}$. □

The direction ($\Leftarrow$) holds for every regime, black-box or white-box, and uses admissibility only through the scheduler-independence of the report. Theorem 8.9 is a restatement, not a new technique: its content is in combination with Theorem 8.5.

Corollary 8.10 (The restatement)**.** *Under the hypotheses of Theorem 8.9,*

$$\mathbf{P} \neq \mathbf{NP} \quad \Longleftrightarrow \quad \begin{array}{l} \textit{no admissible regime lets } [\![\Phi]\!](x) \textit{ certify, at} \\ \textit{polynomial cost, that a flat process cannot succeed.} \end{array}$$

Black-box regimes provably cannot (Theorem 8.5), so the remaining content of $\mathbf{P} \neq \mathbf{NP}$*, in this formulation, is that white-box access affords no exploit. A proof of* $\mathbf{P} = \mathbf{NP}$ *would exhibit a laboratory.*

9 The barriers, located

The correspondences in this section are analogies. We record them because they say where in the apparatus each barrier would bite.

Relativisation and black-box access. An argument that applies uniformly to black-box regimes treats the input as an oracle. Theorem 8.5 and Remark 8.7 show that such arguments can separate unconditionally while saying nothing about the unrelativised question. This mirrors the barrier of [5]. A full analog would also need a regime playing the role of the collapsing oracle; a regime with a primitive experiment answering a **PSPACE**-complete question would do, but it is not admissible in our sense, since that experiment does not cost polynomial phlogiston.

Algebrisation and graded aggregate readout. A graded readout of the kind excluded from Definition 8.4 returns an aggregate — a weighted sum over the resolutions of a copy — rather than the one outcome the scheduler chose. Heuristically this resembles querying an extension of the oracle beyond its truth table, the access model of the algebrisation barrier [2], though a weighted sum is not a low-degree extension over a larger field. We conjecture that graded black-box regimes form a second tier that is unconditionally separated and equally uninformative (Conjecture 9.1).

Natural proofs. An argument that certifies hardness through a large, constructive property of a process's full behaviour — its table of outcomes over all interventions — is the analog of a natural proof [21]. Making this precise is open.

Conjecture 9.1. *Theorem 8.5 holds, with a bound exponential in d, for black-box regimes extended by graded readouts that aggregate over the resolutions of each copy, provided grades are computed from interface names.*

10 Discussion

Popper's asymmetry, with a price. "P can succeed" is existential. One experiment verifies it, and no number of failed experiments falsifies it until every resolution has been tried. "P cannot succeed" is universal. One experiment falsifies it, and verifying it means exhausting the resolutions [19]. Corollary 8.10 puts a price on the asymmetry. Black-box laboratories pay exponentially, provably. Whether a white-box laboratory can pay polynomially is **P** versus **NP**.

The mortal scientist. A scientist modelled as a cost-accounted rho computation, observing populations through graded where-clauses, works in a black-box regime equipped with a distribution over schedules. Its evidence of absence is statistical — a tail bound, not a certificate. The randomised analog of Corollary 8.10 would compare such regimes with **BPP** or **AM** rather than **P**, and may be the more tractable version inside cost-accounted rho, because the graded structure already measures how much a failure tells.

Automata learning. Repeated black-box experiments with steerable choice are membership queries. Determinate systems are learnable from membership and equivalence queries in polynomial time [3]. Under cryptographic assumptions, membership queries do not help for nondeter-

ministic automata [4]. This matches the determinate/nondeterminate split of Proposition 4.3 and Theorem 8.5, though the correspondence has not been worked out.

Independence, revisited. If $\mathbf{P} \neq \mathbf{NP}$ were independent of ZFC, ZFC could not rule out a laboratory. In some model with nonstandard integers, a nonstandard $[\![\Phi]\!]$ would exploit a regime whose cost bound is a nonstandard polynomial. The theorem of [7] concerns a different theory, PA with all true Π_1 sentences, and does not apply to ZFC directly. It does indicate how close to real such laboratories would have to be if independence held for theories of that kind. The Boolean-valued models of [10] have random variables as their integers. Grades valued in such an algebra are a natural candidate for the fuzzy rho of those models, and connecting the two is open.

11 Open problems

1. **Compositional normalisers.** Prove Conjecture 4.5. It is the analog of a monotone lower bound, and the most promising source of a provable restricted separation.
2. **Equivariant stamping.** Realise (E3) for general rho processes by a namespace-relative evaluation in which every name the copy synthesises is stamped, or show that no rho context achieves it.
3. **Drop-only congruence.** Prove Conjecture 6.2, making tier (b) an admissible observer class in the sense of [24].
4. **Graded regimes.** Prove Conjecture 9.1, and identify the quantum analog with its $\Omega(2^{d/2})$ bound.
5. **Statistical regimes.** State and prove the randomised analog of Corollary 8.10.
6. **Minimal labels.** Replace enabling contexts by minimal ones, following [11, 24], and check that Proposition 3.3 survives.
7. **Cost hypotheses.** Verify Hypotheses 2.7 and 2.8 for cost-accounted rho as implemented, and determine the cost of probes in the observer extension.
8. **The intermediate tier.** Characterise regimes with positional interventions but no full reading of the term. Is there an access discipline strictly between (b) and (c) that is still provably separated?

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A Status of results

Result	Status	Note
Prop. 3.3	sketched	administrative phases as in [12]
Thm. 3.5	known	(b) via [23]; reduction sketched
Faithfulness remark after Prop. 4.3	proved	
Prop. 3.7	proved	
Thm. 4.2	proved	
Prop. 4.3	proved	unconditional
Conj. 4.5	conjecture	open problem 1
Lem. 5.1	proved	syntactic
Ex. 5.2	illustrative	formal content in Thm. 8.5
Conj. 6.2	conjecture	open problem 3
Lem. 7.2, 7.3	proved	
Ex. 7.4	proved	
Prop. 7.6	proved	given (E1)–(E3)
Lem. 8.2	proved	relative to Hyp. 2.8 and Def. 8.1
Thm. 8.5, Cor. 8.6	proved	in the model of Def. 8.4; copy count unconditional, cost via Hyp. 2.8
Thm. 8.9, Cor. 8.10	proved	relative to Hyp. 2.7, 2.8 and probe cost; a restatement
§9	analogy	not theorems
Conj. 9.1	conjecture	open problem 4